Rishi Raj, PhD

Scientific Computing | C++/Python | HPC | Data Compression

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PhD in theoretical and mathematical physics from Sorbonne University, now building lossless C++ compression for the CMS experiment at CERN. Three peer-reviewed papers; skilled in numerical algorithms and high-performance C++/Python, plus FPGA/GPU training. Seeking roles in scientific software, HPC, ML and quantum computing.

RESEARCH EXPERIENCE

CERN | Research Placement Oct 2026 – present

CMS experiment, Next Generation Triggers project | Shrinking raw detector data for the upgraded LHC

- Designed a lossless C++ data format for the pixel detector: 45% smaller than the current CMS format with the same compressor, bit-exact on every test event, and fast (10 ms to encode, 5 ms to decode an event).
- Benchmarked 41 compression settings (ZSTD, LZMA, LZ4, Meta's OpenZL) with held-out test data: trained OpenZL compressors produced files 13% smaller than ZSTD at the same compression speed (105 MB/s).
- Measured lossy options through the full particle-tracking pipeline on 2,000 simulated collisions: data cut to about a third of the original size for a 0.27-point loss in tracking efficiency.
- Trained neural-network probability models (on an H100 GPU) for compressing detector hits, and evaluated compact formats for the new calorimeter, within a few percent of the theoretical minimum size. Technical note in preparation.

Sorbonne University | Doctoral Researcher, Theoretical & Mathematical Physics Oct 2023 – Jun 2026

- Computed exact counts of black-hole bound states in string theory, reducing a hard quantum problem to finite integrals, with analytic and numerical checks (JHEP 2026).
- Mapped which particle states are stable in a string-theory model across three regimes; built the computations as 80+ Mathematica notebooks and a reusable package, validated numerically (Ann. Henri Poincaré 2026).
- Extending the approach to a related model (paper in preparation); presented at conferences and seminars.

SELECTED PROJECTS

Research Paper | Machine Learning on Black-Hole Simulations (Phys. Rev. D, 2025) 2023 – 2025

- Simulated 10,000 far-from-equilibrium black holes from random initial states by solving Einstein's equations numerically (spectral methods, 4th-order time stepping), recording a boundary signal and the horizon size.
- Trained neural networks to predict horizon size from the signal alone: test $R^2 = 0.998$ on the full 299-point series and 0.93–0.96 from just 10 engineered features; 70/15/15 split, five-fold cross-validation.

Master's Thesis | C++ Simulation of a Matrix Model of Black Holes (IIT Madras) 2022 – 2023

- Wrote a templated C++20 simulator (Boost.Odeint, 6th-order symplectic integrator) for up to 1,440 coupled differential equations; 100–200× faster than the Mathematica prototype.
- Ran week-long batch campaigns on a 64+ core cluster; showed the system is chaotic and settles close to random-matrix statistics, independent of where it starts.

TECHNICAL SKILLS

Programming: C, C++17/20, Python (NumPy, SciPy, pandas), Mathematica, SQL, Bash.
Tools and systems: Linux, Git, CMake, Docker, pytest, ROOT, CMSSW, PBS batch clusters.
Numerics: ODE/PDE solvers, spectral methods, linear algebra, optimisation, statistics, information theory.
Machine learning: neural networks (MLPs), gradient boosting, cross-validation; PyTorch, scikit-learn, XGBoost.
Spoken languages: English (fluent), French (A1), German (A1).

EDUCATION AND TRAINING

PhD in Theoretical and Mathematical Physics | Sorbonne University 2023 – 2026

Thesis: *BPS Black Holes and Donaldson–Thomas Invariants*; defended 10 June 2026.

CERN STEAM Academy | Scientific Computing, Hardware and ML (CERN, Geneva) Jun – Aug 2026

Ten-week on-site programme (about 470 hours); overall grade 5.5/6, top 6 of 24 participants.
Performance computing: modern C++/SIMD, OpenMP, CUDA, cache-aware layouts, RDMA/InfiniBand, MPI, storage.
Hardware: FPGAs in VHDL, high-level synthesis (Vitis HLS), Zynq SoCs, ML on FPGAs (hls4ml), optical links.
Machine learning: deep learning, transformers, anomaly detection, explainability, generative models.

BS (Hons) and MS in Theoretical Physics | IIT Madras 2018 – 2023

Minor in Mathematics. MS GPA 9.59/10; overall GPA 9.18/10.

PUBLICATIONS, AWARDS AND ACTIVITIES

Publications: B. Pioline, R. Raj, JHEP 03 (2026) 179; B. Le Floch, B. Pioline, R. Raj, Ann. Henri Poincaré (2026); V. Jejjala, S. Mondkar, A. Mukhopadhyay, R. Raj, Phys. Rev. D 111, 026016 (2025).
Awards: COST research-visit grant, University of Geneva (Feb 2026); Charpak Lab Fellowship (Campus France, 2023); 63rd and 64th Institute Day Prizes (IIT Madras); KVPY and NIUS science fellowships (2018–2023).
Activities: Patatrack GPU-software hackathon, CERN (Oct 2026); 4EU+ PhD School on ML in Physics, Copenhagen (Jun 2025); ROOT open-source Fixathon, CERN (Feb 2024).